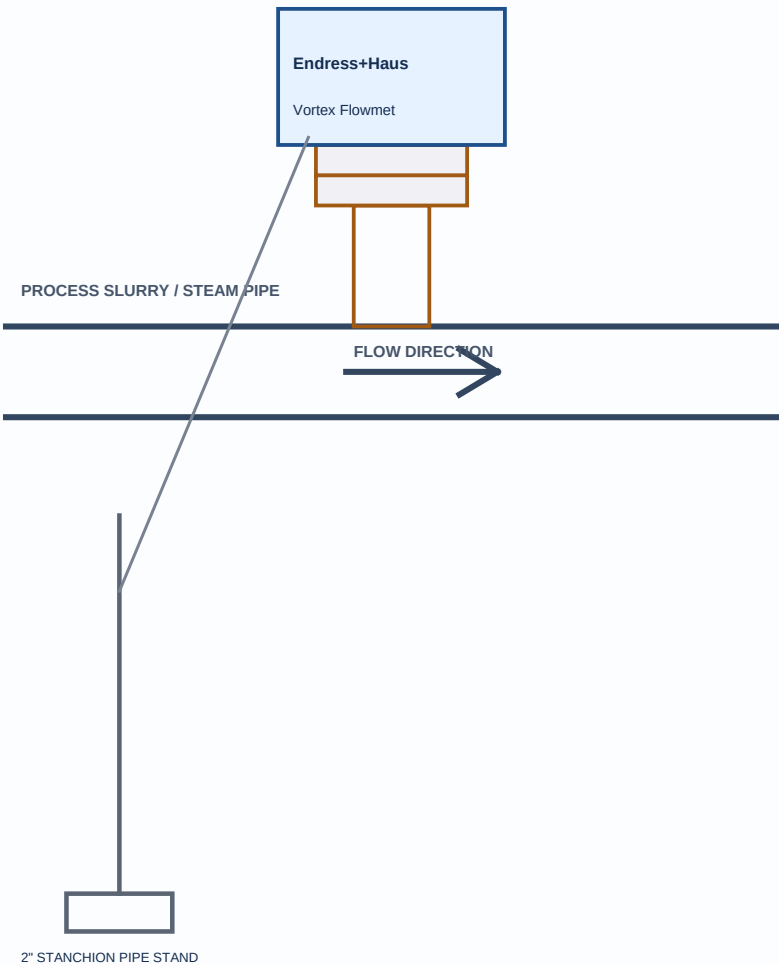


TEMPLATE TYP-08: VORTEX FLOWMETER

TYPICAL TAGS: FT-60101 / FT-60202

SERVICE: High-Pressure Saturated Steam Flow Measurement | PLC I/O: 1756-IF16 (Compensated Mass Flow Steam)

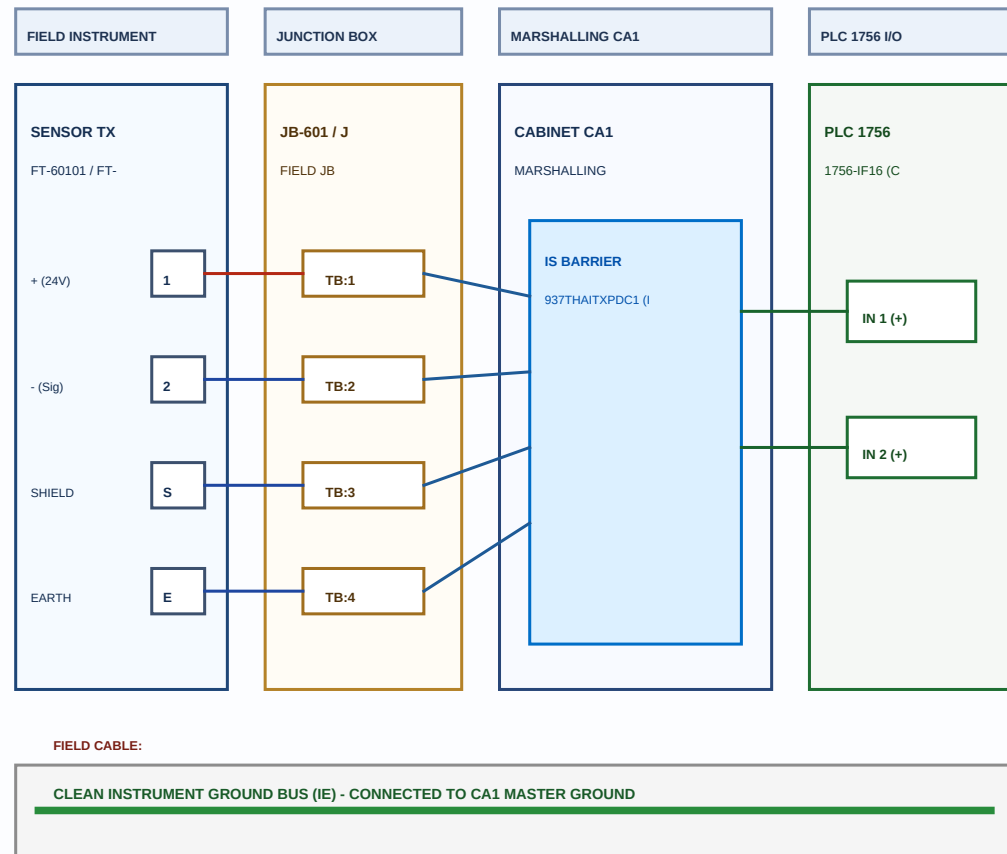
SECTION A: PROCESS PIPING & MECHANICAL HOOK-UP



MECHANICAL HOOK-UP SPECIFICATION

Process Connection:	DN80/100 (3"/4") ASME Cl.300..
Mounting Type:	Direct Flanged / 2" Pipe Stand
Enclosure Rating:	Dual-Compartment IP67 Die-ca..
Hazardous Class:	ATEX II 1/2G Ex ia IIC T6 Ga..
Process Medium:	Starch Slurry / Saturated St..
Ambient Temp Range:	-20°C to +60°C Operating

SECTION B: ELECTRICAL LOOP WIRING SCHEMATIC (FIELD TO PLC)



ELECTRICAL & SIGNAL LOOP SPECIFICATIONS

Signal Protocol:	4-20mA HART Steam Flow (0-10,000 kg/h)
Power Supply:	24V DC via 1756-IF16 Analog Channel
Field Cable Type:	1Px1.5mm ² Twisted Shielded (High Temp ..
Destination JB:	JB-601 / JB-602
Marshalling Cabinet:	CA1 / Master Control Panel (Bay M1..M4)
Barrier / Isolation:	937THAITXPDC1 (IS Loop Powered Repeater)
PLC Module:	1756-IF16 (Compensated Mass Flow Steam)

SECTION C: BILL OF MATERIALS (BOM) & NOTES

MANUFACTURER: ENDRESS+HAUSER

MODEL: Prowirl F 200 (7F2C80 / 1H with Temp Sensor)

SERVICE: High-Pressure Saturated Steam Flow Measurement

ITEM	MATERIAL / COMPONENT DESCRIPTION	SPEC / MODEL	QTY
1	Prowirl F 200 Flanged Vortex M E+H SS316 Cast Body w/ DSC Senso		1 EA
2	Counter-Flanges DN80 Cl.300 RF ASTM A105 / A182 F316 Flanges		2 EA
3	Spiral Wound Gaskets Cl.300 Gr Graphite Filled SS316 Rings		2 EA
4	Cable Gland M20x1.5 Ex d/e Hig Brass Nickel-Plated Gland		1 EA
5	Pipeline Insulation Spacer & S Thermal Extension Neck Jacket		1 SET

ENGINEERING INSTALLATION NOTES

- Straight piping: min. 15D upstream and 5D downstream from flow disturbers (valves, reducers).
- Insulate pipe line up to meter neck, but leave transmitter electronics housing uninsulated.
- Verify that internal integrated Pt100 temperature compensation is activated in HART configuration.

INGREDION (THAILAND) CO., LTD.
KALASIN STARCH PLANT - JET COOKER PROJECTAEC INDUSTRIAL ENGINEERING
SYSTEMS INTEGRATION & AUTOMATIONVORTEX STEAM FLOWMETER HOOK-UP & LOOP DIAGRAM
INSTRUMENT HOOK-UP & TYPICAL LOOP DIAGRAMDWG NO: KAL-JC-TYP-008
REV: 3.6 DATE: 2026-09-08DRAWN: AEC-ENG
CHECKED: LEAD-PESTATUS: APPROVED
SCALE: N.T.S.PROJECT NO:
R5THS00138-JCSHEET NO:
10 / 25